

TECHNICAL RESEARCH REPORT · CORRECTED P0

Four days. Evidence-led alerts.

A tightly scoped four-person build: vendor-neutral registry, reviewable ANPR sightings, authorised watchlist alerts, and inferred GIS movement — with departmental recording left in place.

Sources: official brief, 31-slide team plan (31 August 2026), local papers folder after removing a misfiled dropsonde paper, and the four-day P0 correction.

Date: 1 September 2026 · Status: research findings, not a working prototype claim.

Execution: four people, four days. Decisive Day-1 question: government-feed access, not model selection.

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1. Executive summary and corrections

Twenty-six Gujarat departments run independent CCTV estates, scattered up to about 1,000 km, with different vendors, VMS platforms, storage, and retention. The scoring object is narrower: onboard about 50 heterogeneous government feeds, recognise a designated registration number, reconstruct timestamped GIS movement, and raise real-time watchlist alerts from a working backend.

This report remains research rather than a measured prototype. The architecture recommendation is unchanged in kind: a phased hybrid; departmental video retained locally; vehicle ANPR as the primary scoring path; raw OCR plus evidence retention; exact matching and human review; GIS sightings with inferred — not proven — links; and no Kafka, Kubernetes, or full central VMS during the prototype. Government-feed access and hardware benchmarking remain the critical unknowns.

Proceed only as a tightly scoped, four-person P0 build. The strongest submission story is not “we built a statewide VMS.” It is: we onboard heterogeneous government feeds into a vendor-neutral registry, continuously generate reviewable ANPR sightings, correlate them with an authorised watchlist, and show timestamped evidence and observed movement on GIS, while retaining existing departmental recording infrastructure.

1.1 Corrections to the previous draft

Issue	Correction
Schedule	The 14-day / four-person plan in the team deck (slide 22) is not usable for this deadline. Four days needs a much smaller P0 and parallel ownership. Compress the deck's Day 1–3 critical path (slide 26) into the first two days.
Fifty-camera scope	“Two source types” is useful heterogeneity evidence. It does not replace onboarding approximately 50 supplied cameras. Every accessible camera must appear in the ledger. Actual analytics coverage must be reported.
Awiros recogniser	Candidate, not locked. 98.42% overall / 96.91% two-row at ~5 ms/crop on an RTX 3090 is OCR on pre-cropped plates. The report does not state the held-out validation-set size for those scores, provides no external Gujarat-feed test, and does not measure the full detector-to-alert pipeline. Not expected portal-feed accuracy.
Misfiled paper	2023_CityScale_Vehicle_Trajectories_SciData.pdf is a meteorological dropsonde dataset (ACTIVATE). Removed from the CCTV evidence pack. Do not cite it as city-scale vehicle research.
Publication year	The modular YOLO / PaddleOCR file is a Sensors 2026 article (26, 2785), despite the 2025_ filename and the earlier “Sensors 2025” label.
External claims	ONVIF versions, model licences, Hugging Face model-card licences, MoRTH/NAPIX access, DPDP interpretation, and patent status are verification tasks, not settled facts from the local papers.
Patents	Do not make a patent the implementation specification. Implement a simple independently designed voting / consistency method. Obtain IP review before any patent-derived production commitment.
2 FPS	Starting hypothesis only. Calibrate sampling from vehicle speed, plate pixel width, dwell time, blur, and measured passage recall.

2. Problem statement and evaluation test

2.1 Official problem

Departments operate standalone camera ecosystems. Some store in the cloud, some locally; retention is 7 or 15+ days. The brief also names VAHAN, SARATHI, eGujCop/CCTNS, AFIS, and NAFIS. AFIS/NAFIS are fingerprint systems and are out of the ANPR path. Live national-database access is not a four-day deliverable.

2.2 Live test (the scoring object)

Requirement	Meaning for the four-day build
Onboard ~50 heterogeneous cameras	Every accessible camera in the ledger (connected, blocked, or deferred, with reason). Two source types prove heterogeneity; they do not replace the ~50-camera scope. Report <code>analytics_active_count</code> versus onboarded count.
Integrate live or recorded feeds	Working backend on the actual host. Mock UIs are disallowed. Recorded input must be labelled.
Designated registration number	Identify, timestamp, GIS history, searchable events from persisted sightings.
Watchlist cross-reference	Representative (synthetic) list is allowed. Exact match auto-alerts. Fuzzy is review-queue only and droppable.
Own-feed demo 2–3 min	Onboard, then analytics, match, and alert.
Government-feed demo	Separate deliverable plus plate/timestamp output report. Blocked if Day-1 decode fails.
Scale narrative to ~80,000	Roadmap only. Not a fake 80k-stream benchmark from one host.

Official submission pack: solution PPT/PDF, high-level design, own-feed video, government-feed video and output report, optional hosted URL and repository.

3. Architecture decision and scope locks

3.1 Phased hybrid (Model 5)

Model	Role in P0	Trade-off
1 Registry + GIS	Always included: inventory, ownership, health, gap analysis.	Inventory alone cannot detect vehicles.
2 Direct integration	First working feed-to-alert path (RTSP / ONVIF / vendor API).	Needs network access; loads the source.
3 VMS federation	Out of four-day scope unless an API is already in hand.	Vendor licence dependence.
4 Central VMS	Not built. Selective incident crops only.	Blanket central recording is costly.
5 Hybrid	Site-by-site path from actual capability.	Needs consistent contracts and ownership.

Target topology: departmental recording stays local, then a regional gateway (adapters, relay, ANPR, durable event queue), then central services (registry/GIS, watchlist, matching, alerts, audit). For the hackathon these layers may be co-located on one host. Do not claim edge or WAN savings if all feeds are pulled centrally.

3.2 Locks that survive the four-day cut

Lock	Decision	Four-day effect
Performance	Live ~50-feed demo	Report actual analytics coverage. Do not fake healthy cameras.
Analytics domain	Vehicle ANPR + watchlist only	No face, no person ReID, no FRS.
ANPR algorithm	Two-stage detector + specialised OCR	Awiros is a candidate, not the locked recogniser.
“Complete route”	Observed sightings + inferred GIS path	Dashed links, not verified roads.
Watchlist match	Exact normalised match + human review	Fuzzy review queue is droppable at midday Day 2.

3.3 Build / drop / defer

Build in four days	Drop if a gate fails	Defer
Registry + GIS ledger of all accessible cameras	Fuzzy matching	Facial recognition
ANPR sightings + exact watchlist + evidence	ByteTrack / WPOD-NET / relay refinements	Visual ReID / Xie travel-time / EASE-MCVT
RBAC, audit, secrets from day one	Bonus features after a missed gate	Live VAHAN / NAPIX / NAFIS; Kafka; Kubernetes; full central VMS

4. Four-day P0 plan and hard checkpoints

The existing deck's 14-day plan for four contributors (slide 22) is planning evidence, not the execution schedule. Four experienced people still cannot deliver that scope in four days. Parallel ownership is mandatory: A integration, B ANPR, C backend, D UI/GIS.

Window	Work (compressed from deck Days 1–3 into Days 1–2)	Owner
Day 1 morning	Decode an authorised government feed on the actual host. Start the camera ledger. Freeze the event schema. Stand up detector + candidate OCR.	A, C, B
Day 1 afternoon	Persist a readable plate as a real sighting (raw OCR, crop, camera, timestamp). Minimal GIS point only if that sighting exists.	B, C, D
Day 2 morning	Exact watchlist match becomes an alert with evidence and GIS on one record. Two source types if possible; continue the ~50-camera ledger.	A–D
Day 2 afternoon	Expand onboarding. Calibrate sampling from dwell time and passage recall. Exact-match only unless the midday gate passed.	A, B, C
Day 3	Required concurrency. Report analytics_active_count versus onboarded count.	A, B
Day 4 to midday	Feature freeze. Record both demos. Output report. HLD and links. Claims must match demonstrated behaviour.	D + all

4.1 Hard checkpoints

When	Fail condition	Immediate action
Midday Day 1	No government feed decoded on the actual host	Escalate access immediately. The mandatory government-feed demonstration is blocked until it works.
End of Day 1	No readable persistent sighting	Stop UI and infrastructure work.
Midday Day 2	No real end-to-end alert	Remove fuzzy matching, advanced tracking, relay refinements, and bonus features.
Midday Day 3	No stable required concurrency	Reduce inference sampling from measured plate dwell time. Disclose actual analytics coverage.
Day 4 midday	—	No new features after this.

Day-1 question. If authorised government feeds cannot be decoded on the actual host during Day 1, the mandatory government-feed demonstration is blocked regardless of how polished the remaining system becomes.

5. Prototype pipeline to build

Licence-clean, real-time chain after the paper filter. Video stays on the analytics path even if nobody has the dashboard open. The UI must not decode 50 full-resolution browser tiles.

1. Ingest RTSP / ONVIF / file
2. MediaMTX — one pull; WebRTC view plus analytics tap (drop relay refinements if midday Day 2 is missed)
3. FFmpeg decode at a **calibrated** analysis FPS — 2 FPS is a hypothesis, not a lock
4. ONNX plate detector (FastALPR YOLOv9-t or exported YOLO)
5. SORT passage grouping; ByteTrack only if needed; sharpest crop; WPOD only if skew is measured
6. Candidate Indian OCR on crops (Awiros fine-tune is the first benchmark, not locked)
7. Independently designed multi-frame consistency vote; keep raw OCR
8. Exact watchlist match becomes an alert; fuzzy goes to a labelled review queue and is droppable
9. PostGIS sighting plus inferred GIS path

Do not ship the Ultralytics package as a networked service pending AGPL verification. Kubernetes, Kafka, and a full central VMS are not four-day proof of scale.

6. Paper-by-paper disposition

Open-access PDFs in papers/ were read against the locks and the four-day cut. The table is the source of truth for what a paper may be used for. Do not upgrade a candidate into a lock.

6.1 Detection, OCR, Indian plates

Paper	Disposition
Laroca et al., IJCNN 2018	Supports video-level evaluation and temporal redundancy. Brazilian results are not Gujarat accuracy.
Silva & Jung, WPOD-NET, ECCV 2018	Supports perspective rectification. Reported complete pipeline was comparatively slow. Add only when skew is a measured failure.
Zherzdev & Gruzdev, LPRNet, 2018	Lightweight but Chinese-focused. Tanwar shows the Indian two-row limitation. Not the OCR path.
Laroca et al., layout-independent YOLO, 2019	Supports explicit one-row / two-row classification and layout-aware validation. Copy the method, not Brazilian syntax.
Tanwar et al., Indian-LPR, 2021	Strongest local open Indian detector-dataset evidence. Four-point warping: 66.3% to 75.6% character accuracy. 75% remains inadequate for deployment.
Indian truck plates, YOLOv7 + PARSeq, 2022	Relevant for difficult commercial / two-line plates. Validation set is only 239 images. Retain as a fallback; do not dismiss completely.
Nadiminti et al., Indian E2E ANPR, 2022	Supports COCO-pretrained detector adaptation rather than transferring a Chinese CCPD model.
Laroca et al., AOLP/CCPD near-duplicates, 2023	Requires video-level or vehicle-level splits. Adjacent frames must never be divided between training and testing.
Sensors 2024, unconstrained LPR	Strong Chinese CCPD results; explicitly acknowledges restricted plate coverage. Not Indian validation.
Awiros ANPR-OCR technical report	Candidate, not locked. 98.42% / 96.91% two-row at ~5 ms/crop (RTX 3090) is OCR on pre-cropped plates. Held-out validation-set size for those scores is not stated; no external Gujarat-feed test; not a full detector-to-alert pipeline. Promising first benchmark, not expected portal-feed accuracy.
Cui et al., PaddleOCR 3.0	Suitable runtime / toolkit. The general pretrained model is not by itself an Indian ANPR solution.
IJARSCT 2025, YOLOv8 + PaddleOCR (Indian)	Supports the broad YOLO to PaddleOCR pipeline. Insufficient quantitative evidence for model selection.
Islas-Yañez et al., Sensors 2026, 26, 2785	Filename and earlier review said 2025; the article is Sensors 2026. Closest published stack shape. Copy pipeline order only.
Korean YOLOv12 + OCR, Sensors 2026	Useful mainly as a warning: tracking errors and integrated-pipeline overhead dominate isolated OCR benchmarks.

6.2 Tracking, GIS, matching

Paper	Disposition
Bewley et al., SORT, 2016	Lightweight fallback for per-camera passage grouping. Detection quality dominates tracking quality. Default tracker for P0.
Zhang et al., ByteTrack, 2021/22	Useful inside one camera for grouping detections and selecting the best crop. Not needed for cross-camera identity. Drop if midday Day 2 is missed.
Xie et al., multi-camera travel-time, 2022	Visual-ReID travel-time system. Future-roadmap material, not necessary for registration-number tracking.
SAE EASE-MCVT, 2025	Supports regional edge processing and central metadata. Visual ReID and its engineering framework are outside the four-day scope.
CRS R48160, ALPR hotlists, 2024	Supports hotlist comparison, real-time alerts, privacy controls, and human review.
Misfiled SciData 2023 PDF (removed)	REMOVED. Meteorological dropsonde dataset (ACTIVATE / western North Atlantic), not city-scale vehicle research. File is in papers/_excluded/.

7. Datasets, syntax, tools

7.1 Data to use

Asset	Role	Do not
Indian-LPR (Tanwar) — 16k images, 4-point	Detector train/val; two-row split is the hard set.	Treat 75.6% as deployment-ready, or still-image mAP as government-feed accuracy.
Awiros weights (candidate)	First OCR benchmark if the model-card licence verifies. Pin hash.	Claim 98% on Gujarat portal feeds. Treat as a locked recogniser.
Indian truck PARSeq set (val n=239)	Fallback for commercial / two-line plates.	Dismiss it completely, or treat n=239 as sufficient validation.
Authorised own-feed video + synthetic watchlist	Mandatory own-feed demo. Plates in the video must be in the list.	Harvest real private GJ plates as “stolen.”
Government portal feeds	Scoring object. Protocols unknown until they decode on the host.	Pretend own-feed satisfies the government-feed test.
Kaggle Gujarat plates (355 images)	GJ font smoke test only.	Train a detector from this alone.

CCPD, AOLP, and UFPR-ALPR are the wrong fonts/layouts for training. UFPR is only a reminder that unconstrained video is much harder than stills. Adjacent frames must never be split between training and testing (Laroca et al., arXiv:2304.04653).

7.2 Indian plate syntax (plausibility filter)

Family	Pattern	Note
Standard (Motor Vehicles Act 1988)	SS DD XX NNNN (series 0–3 letters; I/O unused)	Example: GJ 01 AB 1234 — re-verify cited instruments.
Some states omit leading district 0	SS D XX NNNN	Gujarat / Delhi style
Bharat series (MoRTH 2021)	YY BH ##### XX (I/O unused)	Example: 26BH4567AB — must not be dropped
Gujarat government	GJ 18 G ##### / GJ 18 GJ #####	Not a private series
Two-wheeler two-line	State+RTO on line 1; series+number on line 2	Main OCR failure mode

Invalid syntax: keep the raw read, no auto-alert. Valid syntax + exact watchlist becomes an alert. Do not aggressively substitute O/0 or I/1 on the raw string.

7.3 Tools that belong in the prototype

Tool	Job	Caveat
MediaMTX	One RTSP pull, then WebRTC + analytics tap.	Drop relay refinements if midday Day 2 is missed. Licence is a verification task.
ONVIF Profile T / S / G / M	GetProfiles / GetStreamUri; G playback; M vendor metadata.	Which profiles the supplied cameras implement is a verification task.
FFmpeg	Decode.	Lowering inference FPS does not cut WAN if the source stream still moves.
FastALPR YOLOv9-t ONNX	Candidate plate detector.	Swap OCR independently. Licence re-verify.
PaddleOCR runtime + Awiros weights	Candidate OCR on crops.	Toolkit is not an Indian ANPR solution. Awiros is not locked. HF card licence is a verification task.
SORT (ByteTrack optional)	In-camera passage grouping and best crop.	Not cross-camera identity. Detection quality dominates tracking quality.
FastAPI + PostgreSQL/PostGIS + React/Leaflet	API, registry, GIS, alert queue.	No Kafka. Stop UI work if no Day-1 sighting.
Docker Compose + ONNX Runtime	Prototype deploy and inference.	Not Kubernetes. Not an AGPL Ultralytics service.

8. Matching, GIS, ingest, and evidence

8.1 ANPR worker (per feed)

- Decode at a calibrated analysis FPS. Do not hard-code 2 FPS. Start from 2 FPS only as a hypothesis. Calibrate from vehicle speed, plate pixel width, dwell time, blur, and measured passage recall.
- Detect plates (ONNX). Optional vehicle-first if plates are tiny in 1080p.
- Track with SORT. Passage = track ID + short gap. ByteTrack only if grouping quality needs it.
- Score crop: detector confidence × sharpness × minimum plate width.
- OCR top-K crops with the candidate recogniser. Keep raw strings.
- Independently designed multi-frame character consistency / vote. Not a patent implementation. Keep raw OCR plus the voted plate.
- Normalise (strip space/hyphen, uppercase, NFKC). Syntax flag only.
- Exact active watchlist becomes an alert. Optional near-match and syntax-ok go to a review queue, never rewrite OCR. Drop fuzzy if midday Day 2 is missed.
- Persist every sighting so a designated-vehicle GIS search works even without a prior watchlist hit.

8.2 Time, GIS, viewing

Store `source_time`, `ingest_time`, and `clock_offset_ms`. Display IST. Flag large drift. GIS: order exact (or designated) plates by corrected UTC, plot camera coordinates, draw dashed inferred links, and flag implausible implied speed. Do not claim a verified road polyline. Absence of a sighting is not proof of absence.

Viewing path: one source pull, relay to clients, tiles on demand. Analytics path: process all required accessible feeds continuously. Report `analytics_active_count` versus onboarded count. Under backpressure, drop analysis frames; never mark a camera healthy if analytics stopped.

8.3 Alert workflow (operational analogue: CRS R48160)

US Congressional Research Service R48160 describes the law-enforcement pattern: read plate, compare to a hotlist, raise a real-time alert, then officer review. That is the hackathon test. States: New, acknowledged, confirmed / rejected. No automatic enforcement. Human review is mandatory, not optional polish.

8.4 Minimum event schema

Record	Minimum content
Camera	ID, department, GIS, source, capabilities, health (reported vs probed). Every accessible camera, including blocked ones with reason. Secrets referenced, never exported.
Sighting	ID, camera, event time, raw and normalised plate, confidence, model version, evidence reference.
Watchlist	Record ID, plate, purpose, priority, validity, authority, version, expiry.
Alert	Sighting IDs, watchlist ID, match type (exact / fuzzy), status, unique dedup key.
Evidence / audit	Crop or clip URI, hash, actor, action, time, retention, export log.

Every event also carries `trace_id`, `schema_version`, `source_time`, `ingest_time`, `event_id`. A file hash detects byte changes; it does not prove legal admissibility.

9. Patents, policy, and verification tasks

9.1 Patents — cite; do not implement as spec

Multi-frame OCR consensus is useful. Implement a simple independently designed voting / consistency method. Obtain IP review before making patent-derived functionality a production commitment. Patent legal status (filed, granted, expired, applicable jurisdiction) is itself a verification task.

Patent	Idea	Our posture
US20230394850A1	Vote characters across frames.	Cite as related art. Independent consistency method only.
US20200074211A1	ALPR + hotlist, then real-time LE alert.	Same shape as the live test; cite, do not copy claims.
US20240355130A1	Match a list by character overlap.	Fuzzy review queue only. Droppable at midday Day 2.
US10839303B2 (SAP)	Rewrite a bad read from neighbouring cameras.	Cite. Do not auto-correct identity.
US20240265704A1	VMS + GIS field of view / coverage.	Model 1 gap-analysis citation.
KR101451115B1	ONVIF + non-ONVIF ingest.	Adapter pattern citation.

9.2 Policy (engineering notes, not a legal opinion)

DPDP interpretation, MoRTH/NAPIX access, and related instruments were asserted in the earlier literature review. They are not established by the local paper collection. Treat them as verification tasks. Do not scrape VAHAN. Do not fake a live national-database integration. Prototype retention remains: short rolling buffer of all reads; persist watchlist hits, operator-confirmed events, and audit.

9.3 External claims that must be re-verified

Claim class	Why it is not settled
ONVIF versions	Profile T/G/M text on onvif.org is not proof that supplied Gujarat cameras implement those profiles.
Model licences	PaddleOCR, FastALPR, Ultralytics AGPL, MediaMTX — re-check the current licence files.
Hugging Face model-card licences	Awiros and any other weights. Pin hash only after the card is re-read.
MoRTH / NAPIX access	Police data-sharing circulars do not grant this team access.
DPDP interpretation	Requires legal review, not a literature-review assertion.
Patent status	Google Patents hits are not a freedom-to-operate opinion.

10. Remaining gaps and the Day-1 question

Research cannot close access, hardware, or organiser questions. The workspace still contains documentation and papers only — no code, schema, or prototype.

Gap	Why it blocks
Portal feed protocols on the actual host	RTSP vs player vs files decides the first adapter. Own-feed does not satisfy the government-feed test. Escalate at midday Day 1 if nothing decodes.
How many of ~50 cameras actually decode	Every accessible camera still belongs in the ledger. Analytics coverage must be the measured number, not 50 by assertion.
Named GPU / host throughput	Awiros ~5 ms/crop is OCR-only on an RTX 3090, not this host, and not the full pipeline. Do not promise one GPU without a soak.
Deadline is four days, not 14	P0 only. Parallel ownership. Feature freeze at Day 4 midday.
Awiros as a government-submission dependency	Candidate. Licence card, hash, and Gujarat-feed accuracy are all unverified.
Camera clock / NTP	Wrong order of GIS sightings if ignored.

Illustrative (unmeasured) capacity from the team deck, for HLD honesty only: at 2 Mb/s per camera, 50 cameras about 100 Mb/s and 1.08 TB/day; 80,000 cameras about 160 Gb/s and 1.73 PB/day. The hybrid model does not duplicate that centrally. 50 cameras × 2 analysis FPS = 100 FPS is arithmetic on a hypothesis, not a design lock.

11. Source index

11.1 Project documents

- docs/GUJARAT POLICE HACKATHON.md — official brief, models 1–5, test case, deliverables, evaluation.
- docs/Gujarat_CCTV_Hybrid_Team_Plan.pptx — 31-slide team plan, 31 August 2026 (14-day scenario on slide 22; original 72-hour path on slide 26).
- docs/FOUR_DAY_P0_PLAN.md — compressed four-person execution plan and hard checkpoints.
- docs/PHASE2_LITERATURE_REVIEW.md — corrected literature list with paper-by-paper disposition.
- papers/ — local PDFs after removing the misfiled dropsonde paper to papers/_excluded/.

11.2 Core local PDFs (keep in the evidence pack)

File in papers/	Cite as
2025_Awiros_ANPR_OCR_TechnicalReport.pdf	Awiros, Data-Intelligent ANPR (Indian PP-OCRv5). Candidate OCR benchmark only.
2021_Tanwar_Indian_LPR_in_the_wild.pdf	Tanwar, Tiwari, Chowdhry, arXiv:2111.06054
2025_Cui_PaddleOCR_3.0.pdf	Cui et al., PaddleOCR 3.0, arXiv:2507.05595
2018_Laroca_YOLO_ALPR_IJCNN.pdf	Laroca et al., IJCNN 2018, arXiv:1802.09567
2019_Laroca_LayoutIndependent_YOLO_ALPR.pdf	Laroca et al., arXiv:1909.01754
2018_Silva_Jung_WPOD-NET_ECCV.pdf	Silva & Jung, ECCV 2018 — optional if skew is measured
2022_Nadiminti_Indian_E2E_ANPR.pdf	Nadiminti, Gaur, Bhardwaj, arXiv:2207.06657
2025_PMC_YOLO_PaddleOCR_Modular.pdf	Islas-Yañez et al., Sensors 2026, 26(9), 2785 — not 2025
2016_Bewley_SORT.pdf	Bewley et al., SORT, 2016 — default in-camera tracker
2021_Zhang_ByteTrack.pdf	Zhang et al., ECCV 2022, arXiv:2110.06864 — optional in-camera only
2024_CRS_R48160_ALPR_Hotlists.pdf	CRS Report R48160, 19 August 2024
2022_Indian_Truck_Plates_YOLOv7_PARSeq.pdf	Fallback for commercial / two-line plates (val n=239)

Do not cite papers/_excluded/2023_CityScale_Vehicle_Trajectories_SciData.pdf. It is Vömel et al., Scientific Data (2023) 10:753, dropsonde observations during ACTIVATE.

Limits of this report. No source establishes that the proposed prototype already works. No formal legal compliance assessment, live government-system connection, hardware benchmark, supplier quote, IP opinion, or submission has been performed. Numerical costs and capacity figures in the team deck are illustrative. Claims in any PPT or HLD must match demonstrated behaviour. The four-day plan is a scope cut, not a promise that the live test is already feasible.